# Wet and dry thermometers

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The other day in science, we did this weird experiment – if that's what you call it. We had two thermometers. And around the bulb of one of them we put wet tissue paper. Then we hung them both off a clamp on a retort stand. Every minute we had to read the temperature on each thermometer.

Both of them started at 20°C. A minute later, the dry thermometer was still 20°C, but the wet one had dropped to 18°C. Next time the wet thermometer had dropped even more to 15°C, while the dry one was still the same as before. At three minutes the dry temperature was 20°C again and the wet one still lower at 11°C.

We missed the temperature of the dry thermometer at four minutes, but the wet one was 10°C. At five minutes the dry temperature was again 20°C and the wet one had gone up to 12°C. The teacher told us all to record the results in a table in our books and check the temperatures again five minutes later. At ten minutes both temperatures were 20°C and we noticed that the tissue was completely dry.

**1** Draw and label a scientific diagram of the experiment using a pencil and a ruler.

**2** Record the measurements of the experiment in the table below. Don't forget to give each column a title and a unit.

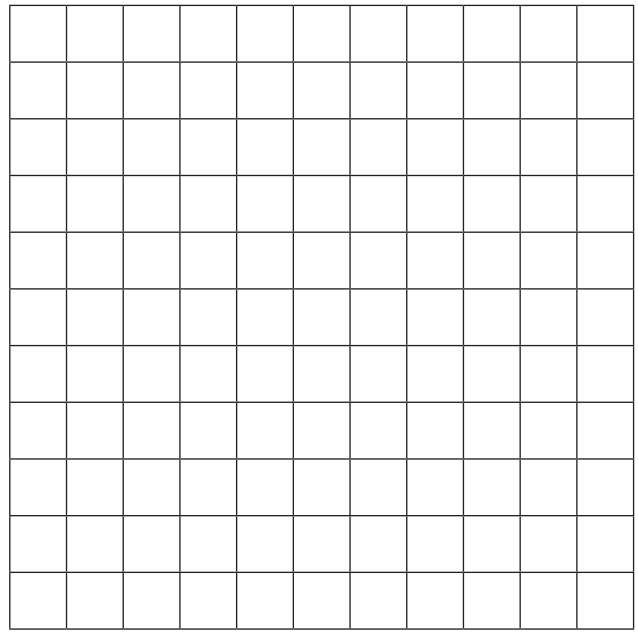
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**3** Label the graph below with a heading and a title and units for each axis. Then plot the information from your table into the graph. Use a blue pen for the dry thermometer and a red pen for the wet thermometer and include a key. Finally, draw a smooth line connecting the blue marks in blue and the red marks in red.



**4** Describe the shape of the graph for each thermometer.

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**5** The wet thermometer started with a wet tissue wrapped around its bulb but ended up with the tissue being dry.

**a** When was the tissue dry for the first time?_____

**b** What happened to the water of the wet tissue?_____

If we want to explain why the wet tissue became cooler as it dried, we need to look at the particles in the water. That is, we are using the particle model of matter to explain this.

**6** **a** How fast are the particles in the liquid water moving?_____

**b** How fast are the particles in the gaseous water moving?_____

**c** Where did the particles get this extra kinetic energy from? _____

**d** If the surrounding, including the thermometer, loses heat energy, what happens to its temperature?

_____

**7** Explain how perspiring cools you on a hot day.

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**8** Suggest why perspiring works well in Alice Springs but not so well in Brisbane.

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